From force diagrams to motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a force diagram to predict acceleration and velocity change.

SUCCESS CRITERIA - I can...

1. Write a force sum along each chosen axis.
2. Connect net force and mass to acceleration.
3. Use velocity direction to decide the initial speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “parallel” and “perpendicular” to explain your model or result. Sentence starters are optional.

parallel: along the same direction as a line or surface

perpendicular: at a right angle, 90°, to a line or surface

component: the part of a vector along one chosen axis

01 Warm-up: bring back an earlier idea

A 3 kg object has 22 N right and 7 N left.
Draw the forces and find net force and acceleration.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

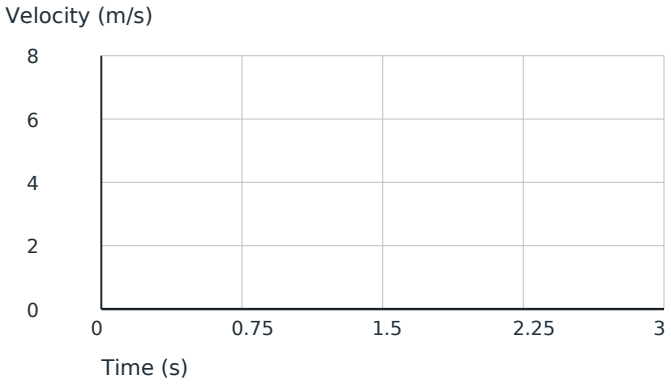
02 Worked example: explain the model

Reference: $a_x = \Sigma F_x/m$; $a_y = \Sigma F_y/m$.
Choose axes first. CPA ramp problems supply the along-ramp gravity component; no trigonometry is required.

Worked reasoning: A 2 kg cart initially at rest has a constant 4 N net force right for 3 s. Acceleration is 2 m/s²; final speed is 6 m/s right.

Your turn: Draw a force diagram and a velocity-time graph that agree.
Mark where mass enters the reasoning.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Horizontal case

The diagram shows forces on a 5 kg box.
Find net horizontal force and acceleration.

Sentence starter: I found ... [unit]. This value means ... in this situation.



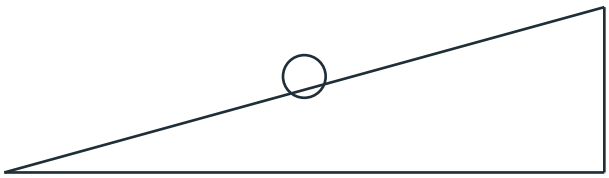
The dot represents the object. Mass: 5 kg

Marzano check - circle: 1 2 3

04 Use a given ramp component

A 2 kg block slides down a ramp. Gravity's down-ramp component is 10 N; friction is 4 N up the ramp.
Find net force and acceleration along the ramp.

Sentence starter: I found ... [unit]. This value means ... in this situation.



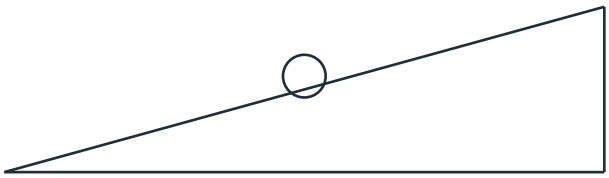
10 N down; 4 N up. The marker represents the object on the sloping surface.

Marzano check - circle: 1 2 3

05 Moving up, accelerating down

A block slides up a rough ramp after the push ends.
Draw its velocity separately from the along-ramp forces. Gravity's along-ramp component and friction both point down the ramp.
Explain what happens to the block's speed.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .



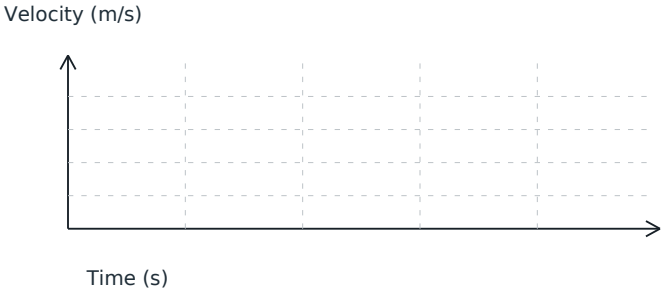
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A velocity graph is constant at -5 m/s for 7 s.
Find displacement. A second object has components 9 m/s east and 4 m/s south; sketch its direction.
Sketch the velocity-time model; label the given values.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 3 kg cart moves left. It has 15 N right and 6 N left. **Find** acceleration. Does its leftward speed initially increase or decrease?

Sentence starter: I found ... [unit]. This value means ... in this situation.



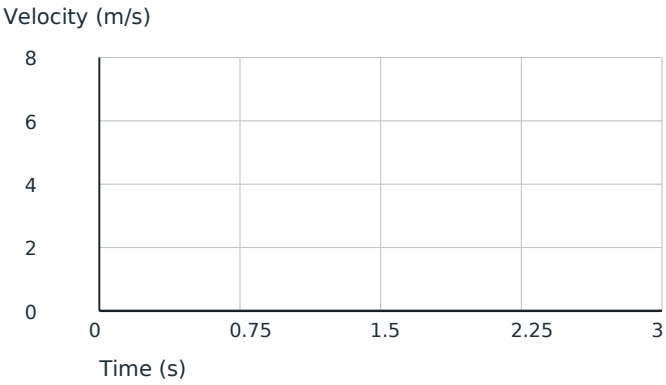
3 kg cart; current velocity is leftward.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Keep the 4 N net force and 3 s time but double the mass. **Compare** the two velocity-time graphs and final speeds.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3